

Luong Hai Long

Computer Vision / Machine Learning / Network Systems Intern | Python | C/C++ | PyTorch | TCP/IP
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Professional Summary

Final-year Electronics and Telecommunications Engineering student at VNUHCM - University of Science, seeking internship roles in computer vision, machine learning, and network/communications systems. Builds Python/C++ CV prototypes and hardware-aware networked systems with PyTorch, OpenCV, YOLO/OCR, TCP/IP, BLE, STM32/ESP32, and Cyclone V SoC.

Technical Skills

Programming languages: Python, C, C++, MATLAB, VHDL, Verilog, Lua, MoonScript.
Computer vision and machine learning: PyTorch, YOLO/Ultralytics, OpenCV, FastALPR, OCR, object detection, image/video inference, Kaggle, IPYNB notebooks.
Networks and communications: TCP/IP, Ethernet, IPv4 subnetting, VLAN, OSPFv2, NAT, DNS, HTTP, DHCP, UART, BLE, BER analysis, AWGN, OFDM, MIMO.
Embedded systems and tools: STM32, ESP32, EFR32xG21, embedded Linux, RTOS concepts, I2C, SPI, LDMA, CMSIS/emlib, Simplicity Studio, Quartus Prime, Git, GitHub, Git LFS, SQLite, Tkinter, FFmpeg, C/Win32 API.

Education

VNUHCM - University of Science Sep 2022 – Expected Sep 2026
Bachelor of Engineering in Electronics and Telecommunications Engineering
Ho Chi Minh City
High-Quality Program, English-taught; Student ID: 22207056; current GPA: 7.2/10.

- Relevant coursework: Artificial Intelligence, Computer Vision, Object-Oriented Programming, Fundamentals of Computer Networks, Network Technologies, Embedded System Design, FPGA/SoC Design, Digital Signal Processing, Digital and Wireless Communications.
- Selected scores: Introduction to Artificial Intelligence 9.1/10; Object-Oriented Programming final exam 9/10; Fundamentals of Computer Networks 8/10; Network Technologies 8/10; Wireless Communications 8.5/10; Embedded Systems Practice 10/10.

Experience

BTP Holdings Jan 2026 – Apr 2026
R&D Embedded Software Engineer Intern Ho Chi Minh City

- Implemented and tested STM32/ESP32 firmware support across C/C++ HAL modules, RTOS-oriented tasks, UART/Serial diagnostics, TCP/IP, and BLE communication paths.
- Built Win32 diagnostic tooling in C for MCU communication over Serial/UART, supporting board bring-up, debugging, and repeatable validation logs.
- Reviewed USB-C Power Delivery behavior and power-board schematics to align firmware tests with hardware constraints during issue tracking.

Robotics & IoT Club, VNUHCM - University of Science Nov 2023 – Jul 2024
Technical Team Member Ho Chi Minh City

- Assembled and tested student robotics and IoT prototypes; debugged sensor wiring, power checks, C/Lua routines, and firmware behavior during prototype bring-up.

Anh Trung Smartphone Store / Xuan Anh Computer 2023 – 2025
Repair Technician Ho Chi Minh City

- Diagnosed phone, laptop, and desktop PC faults across power, display, storage, thermal, connector, software, and peripheral issues; supported repair, setup, testing, and customer handover.

Projects and Technical Coursework

Vietnamese Automatic License Plate Recognition with YOLO and OCR Jan 2026 – Apr 2026
Team of seven Python, PyTorch, YOLO, OpenCV, FastALPR, OCR, FFmpeg, Tkinter

- Co-developed a Python/Tkinter desktop app for image and video inference with OpenCV, FFmpeg, YOLO, and OCR; co-developed a PlateGate PC LAN whitelist demo with one teammate.
- Maintained repository structure, Git LFS assets, Kaggle/IPYNB training workflow, and release packaging for reviewable project delivery.
- Evaluated the detector at 99.45% mAP at 0.5 IoU, 99.45% precision, and 99.37% recall on the validation set.

SoC Ethernet Controller on Intel DE10-Standard Jan 2026 – Apr 2026
Team of five Cyclone V, embedded Linux, VHDL, C/C++, Python, TCP/IP

- Built a TCP/Ethernet command path where PC/Android clients send short payloads to an embedded Linux daemon on Cyclone V HPS.
- Forwarded validated payloads through the HPS-to-FPGA bridge to PIO-mapped registers driving HEX0-HEX5; final project score: 10/10.

AHT20 BLE Data Acquisition on Silicon Labs EFR32 Oct 2025 – Dec 2025
Course project C, SQLite, EFR32xG21, I2C, SPI, BLE, UART/VCOM, LDMA

- Implemented sensor-to-host data flow: AHT20 readings over I2C, Memory LCD display, BLE advertising, UART/VCOM streaming, and C/SQLite logging on Windows.

Wireless and Network Communications Coursework 2025 – 2026
Coursework and MATLAB labs MATLAB, TCP/IP, Ethernet, OSPFv2, OFDM, MIMO, BER

- Completed network coursework covering OSI/TCP/IP, Ethernet, ARP, IPv4 subnetting, ICMP, TCP/UDP, DNS, HTTP, DHCP, VLAN, STP, OSPFv2, NAT, WLAN, and basic JSON/XML/YAML automation formats.
- Modeled communication-system behavior in MATLAB, including BER analysis, AWGN channels, matched filters, BPSK/QPSK/BFSK, OFDM, MIMO, beamforming, and ZF/MRT precoding.